

ES. SIA $f: \mathbb{N} \rightarrow \mathbb{R}_{>0}$ DEFINITA DA

$$f(n) \stackrel{\text{def}}{=} n + \ln(n) + \ln(n)^2.$$

QUALI DELLE SEGUENTI AFFERMAZIONI E' VERA?

$$f = \Theta(2^{n \ln(n)}), \quad f = \Theta(n), \quad f = \Theta(n^2)$$

$$f = \Theta(2^n), \quad f = \Theta(n^2 \cdot \log_{10}(n)) \quad ?$$

$$- f = \Theta(2^{n \ln(n)}) ?$$

BEH...

$$f(n) \stackrel{\text{def}}{=} n + \ln(n) + \ln(n)^2.$$

$$\text{SE } f = \Theta(2^{n \cdot \ln(n)}) \Rightarrow 2^{n \cdot \ln(n)} = O(f)$$

$$\Rightarrow \exists c > 0 \text{ E } \exists N \in \mathbb{P} \text{ TALI CHE}$$

$$2^{n \cdot \ln(n)} \leq c (n + \ln(n) + \ln(n)^2)$$

$$\text{SE } n > N, \text{ MA } n < n \cdot \ln(n) \text{ SE } n \geq 3,$$

QUINDI

$$2^n \leq c (n + \ln(n) + \ln(n)^2)$$

SE $n \geq 3$, CHE È ASSURDO PERCHÉ

$$\lim_{n \rightarrow +\infty} \frac{2^n}{n + \ln(n) + \ln(n)^2} = +\infty.$$

PERTANTO

$$f \neq \Theta(2^n \ln(n)).$$

$$- f = \Theta(n) ?$$

POICHÉ

$$\lim_{n \rightarrow +\infty} \frac{n}{n + \ln(n) + \ln(n)^2} = 1$$

$$\Rightarrow f \cong n \Rightarrow f = O(n) \text{ e } n = O(f) \Rightarrow$$

$$f = \Theta(n).$$

$$- f = \Theta(n^2) ?$$

$$\text{SE } f = \Theta(n^2) \Rightarrow n^2 = O(f) \Rightarrow \exists c > 0$$

$$\text{e } \exists N > 0 \text{ TALI CHE}$$

$$n^2 \leq c(n + \ln(n) + \ln(n)^2)$$

PER $\forall n > N$, ASSURDO. QUINDI $f \neq \Theta(n^2)$.

$$\rightarrow f = \Theta(2^n)?$$

CIOE'

$$n + \ln(n) + \ln(n)^2 = \Theta(2^n) ?$$

POICHE'

$$\lim_{n \rightarrow +\infty} \frac{2^n}{n + \ln(n) + \ln(n)^2} = +\infty$$

CONCLUDIAMO COME PRIMA CHE $f \neq \Theta(2^n)$.

$$- f = \Theta(n^2 \cdot \log_{10}(n)) \quad ?$$

CIOE`

$$n + \ln(n) + \ln(n)^2 = \Theta(n^2 \cdot \log_{10}(n)) \quad ?$$

POICHE`

$$\lim_{n \rightarrow +\infty} \frac{n^2 \cdot \log_{10}(n)}{n + \ln(n) + \ln(n)^2} = +\infty$$

$$\Rightarrow f \neq \Theta(n^2 \cdot \log_{10}(n)).$$

CONCLUDENDO

$$f = \Theta(n).$$

ES. È VERO CHE

$$n! = o((n+1)!) \quad ?$$

ABBIAMO CHE

$$\lim_{n \rightarrow +\infty} \frac{n!}{(n+1)!} = \lim_{n \rightarrow +\infty} \frac{1}{n+1} = 0$$

$$\Rightarrow n! = o((n+1)!) \Rightarrow n! = O((n+1)!).$$